ONLINE BANKING SYSTEM

Business Analysis Portfolio Case Study

Adveidh Krishna
Financial Analyst
LinkedIn
Version 1.0 | July 2026

Portfolio disclaimer
This is a simulated, self-directed Business Analysis case study created to demonstrate requirements analysis, process modelling, Agile delivery, data thinking, documentation and UAT skills. It does not represent work completed for a real bank or employer.

Contents

- 1. Executive Summary
- 2. Business Context and Problem Statement
- 3. Objectives and Success Measures
- 4. Scope, Assumptions and Constraints
- 5. Stakeholder Analysis
- 6. Current-State and Future-State Processes
- 7. Requirements Catalogue
- 8. Business Rules and Use Cases
- 9. User Stories and Product Backlog
- 10. Data Model and Wireframes
- 11. UAT, Risk and Traceability
- 12. Delivery Roadmap and Lessons Learned
- Appendix: Evidence Package and Interview Talking Points

Evidence produced

- Project overview and business case
- Stakeholder register and power-interest matrix
- AS-IS and TO-BE process models
- Functional and non-functional requirements
- Business rules and use-case model
- User stories, acceptance criteria and prioritised backlog
- Simplified ERD and low-fidelity wireframes
- UAT test cases, risk register and requirements traceability matrix

1. Executive Summary

The fictional bank in this case study relies heavily on branch-based services for routine customer transactions. Customers must travel to a branch, wait for assistance and complete manual paperwork for activities that could be completed securely online. The proposed Online Banking System provides authenticated access to account information, transfers, bill payments, statements, beneficiary management and transaction alerts through web and mobile channels.

As the Business Analyst for this simulated initiative, the work focuses on translating customer and operational needs into structured requirements, clarifying scope, analysing stakeholders, improving the transfer process, documenting business rules, supporting Agile delivery and defining UAT evidence.

Item	Summary
Business need	Give customers secure, convenient and traceable access to everyday banking services.
Primary users	Retail banking customers, support agents, compliance staff and operations teams.
Core value	Reduced branch dependency, faster service, lower manual error risk and stronger transaction visibility.
BA outputs	BRD content, stakeholder analysis, process models, requirements, user stories, backlog, wireframes, UAT and risk register.
Delivery approach	Agile Scrum with prioritised releases, iterative validation and traceability from objective to test case.

Key BA question
How can the bank enable convenient digital transactions without compromising security, regulatory control, auditability or customer trust?

2. Business Context and Problem Statement

Current challenges

- Routine transactions depend on branch opening hours and physical travel.
- Customers face queues and inconsistent service times.
- Paper forms and manual re-entry introduce avoidable errors.
- Customers have limited real-time visibility of transfer status.
- Support teams receive preventable enquiries about balances, statements and transaction confirmation.
- The bank lacks a single digital audit trail across all self-service interactions.

Problem statement

Retail customers need a secure self-service channel for viewing accounts and completing common transactions because the current branch-heavy process is slow, inconvenient and vulnerable to manual error. The bank needs a controlled digital platform that improves customer experience while preserving authentication, fraud prevention, regulatory compliance and end-to-end auditability.

3. Objectives and Success Measures

Objective ID	Business objective	Indicative success measure
OBJ-01	Reduce unnecessary branch visits for routine services.	At least 35% reduction in eligible branch transactions within 12 months.
OBJ-02	Improve transaction convenience and speed.	90% of valid domestic transfers confirmed within 60 seconds.
OBJ-03	Strengthen security and auditability.	100% of sensitive actions authenticated and logged.
OBJ-04	Improve customer transparency.	Real-time status and digital receipt for every submitted transaction.
OBJ-05	Reduce avoidable support demand.	20% reduction in balance, statement and transfer-status enquiries.

4. Scope, Assumptions and Constraints

In scope	Out of scope
Secure registration and login with MFA	Corporate banking and treasury services
Account balances and transaction history	International foreign-exchange dealing
Domestic account-to-account transfers	Credit underwriting and loan origination
Beneficiary management	Branch teller workstation replacement
Bill payments and statement download	Cryptocurrency trading
Email/SMS/app alerts and digital receipts	ATM hardware integration
Audit logs and administrative support views	Full core-banking replacement

Key assumptions

- The bank already operates a stable core banking system with secure APIs or integration services.
- Customers have a registered mobile number or email address.
- Regulatory, security and privacy specialists are available for requirement validation.
- A domestic payment network is available for supported transfer types.

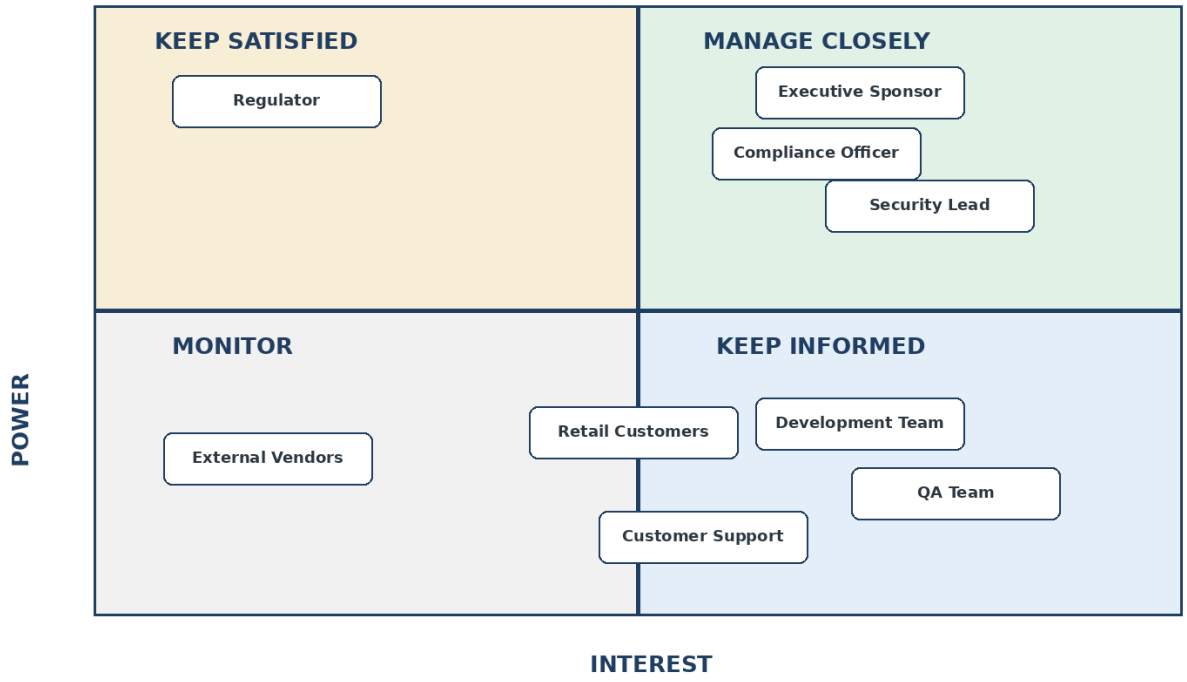
Key constraints

- Compliance with GDPR and applicable banking controls.
- High availability and disaster-recovery expectations.
- Strong authentication for sensitive actions.
- Legacy integration dependencies and limited release windows.

5. Stakeholder Analysis

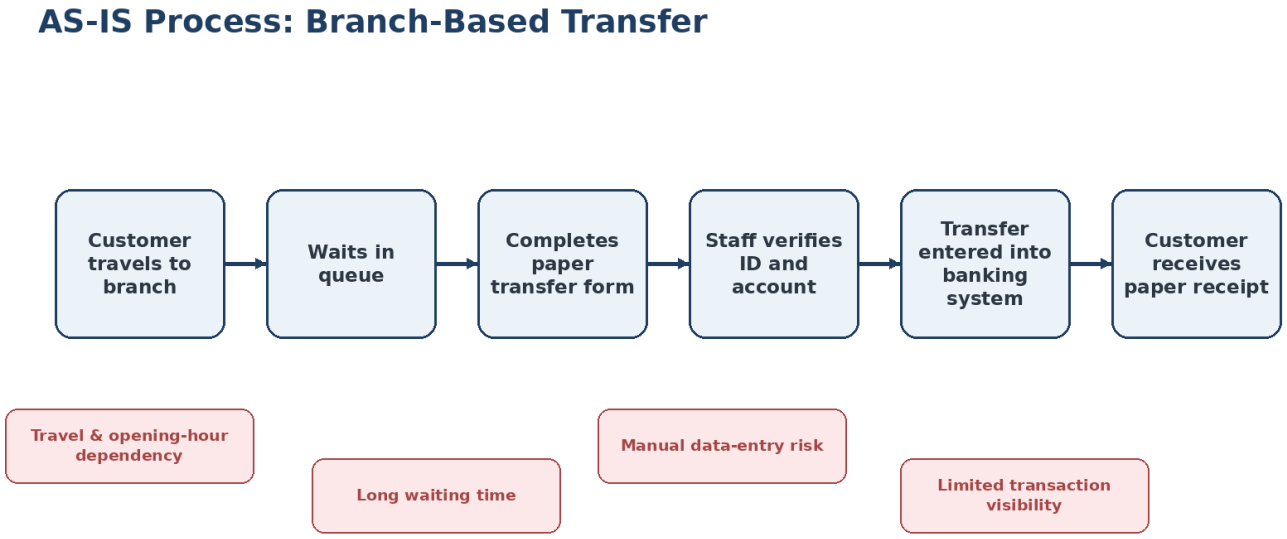
Stakeholder	Interest / need	Influence	Engagement approach
Executive Sponsor	Business value, risk control, delivery progress	High	Monthly steering updates and decision papers
Retail Customers	Ease of use, security, speed, transparency	Medium	Interviews, surveys, prototype testing and UAT
Compliance Officer	Regulatory controls, record keeping, consent	High	Requirements workshops and formal sign-off
Security Lead	Authentication, fraud controls, encryption, monitoring	High	Threat modelling and control reviews
Operations Team	Reliable processing, exception handling, reconciliation	High	Process workshops and scenario walkthroughs
Customer Support	Clear status, searchable history, issue resolution	Medium	Support-process design and knowledge-base review
Development Team	Clear, testable and prioritised requirements	High	Refinement sessions and continuous clarification
QA Team	Traceable acceptance criteria and test data	Medium	Test-planning workshops and defect triage

Stakeholder Power-Interest Matrix



6. Current-State and Future-State Processes

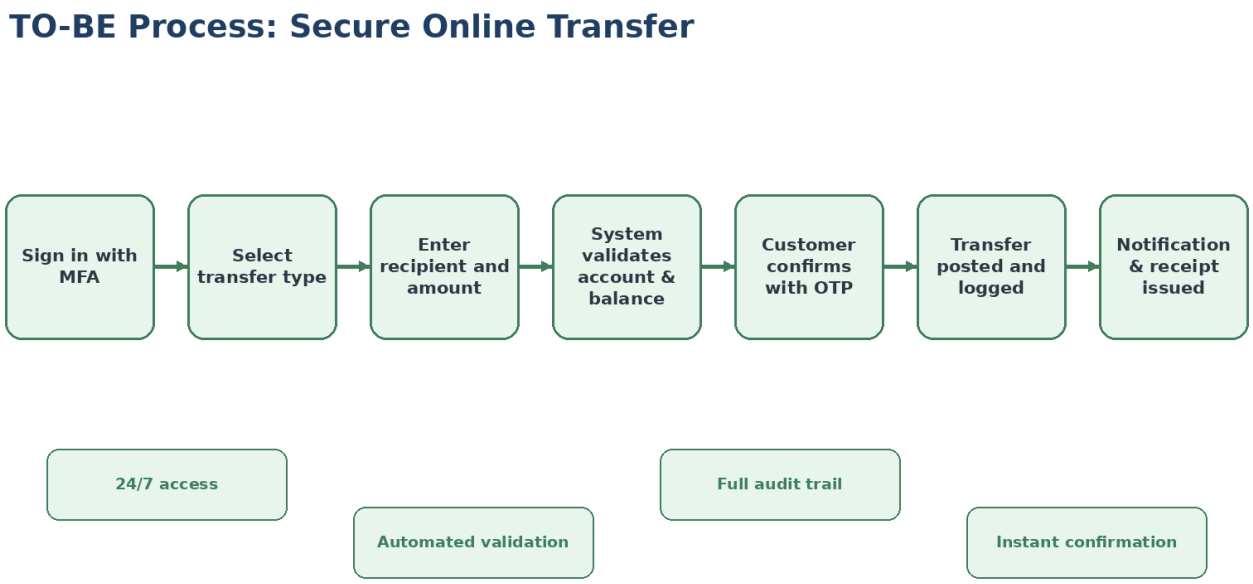
AS-IS: branch-based transfer



Observed pain points

- Travel and queue time increase the customer effort required for routine transactions.
- Paper forms may contain incomplete or incorrect details.
- Staff must manually validate and re-enter information.
- Customers receive limited status information after leaving the branch.

TO-BE: secure online transfer



Expected improvements

- Digital entry removes paper re-keying.
- Automated validation identifies invalid recipients or insufficient funds before submission.
- MFA and OTP controls protect sensitive actions.
- The customer receives an immediate status, reference number and receipt.
- Every step is recorded for support, reconciliation and audit purposes.

7. Requirements Catalogue

Functional requirements

ID	Requirement	MoSCoW
FR-01	The system shall authenticate customers using a username/password and a second factor.	Must
FR-02	The system shall display all active customer accounts and available balances.	Must
FR-03	The system shall display searchable transaction history for at least 24 months.	Should
FR-04	The customer shall be able to add, edit and deactivate beneficiaries.	Must
FR-05	The customer shall be able to transfer funds between own accounts.	Must
FR-06	The customer shall be able to transfer funds to an approved domestic beneficiary.	Must
FR-07	The system shall validate balance, limits, recipient details and account status before confirmation.	Must
FR-08	The customer shall confirm sensitive transfers using OTP or equivalent step-up authentication.	Must
FR-09	The system shall generate a unique transaction reference and digital receipt.	Must
FR-10	The customer shall be able to download account statements in PDF format.	Should
FR-11	The system shall send configurable transaction alerts by email, SMS or app notification.	Should
FR-12	Support users shall be able to search transactions without viewing full authentication secrets.	Could

Non-functional requirements

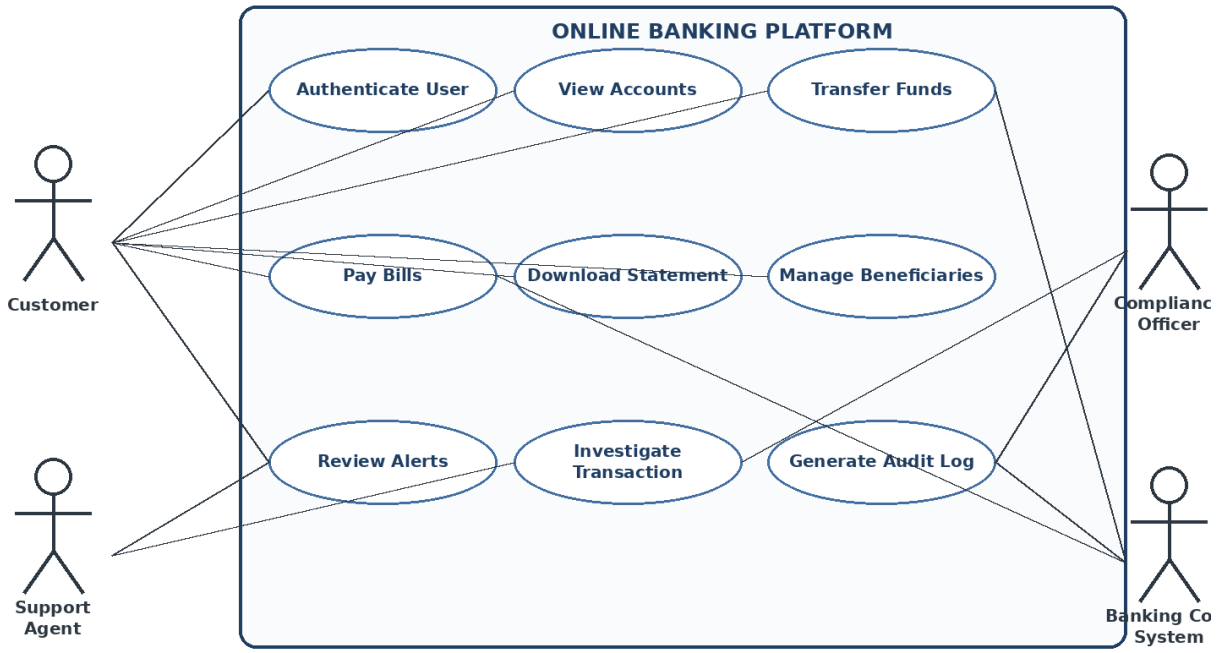
ID	Category	Requirement
NFR-01	Security	Sensitive data shall be encrypted in transit and at rest.
NFR-02	Availability	Customer-facing services shall target 99.9% monthly availability.
NFR-03	Performance	95% of standard pages shall load within 3 seconds under normal load.
NFR-04	Auditability	All authentication, beneficiary and transaction events shall be time-stamped and logged.
NFR-05	Privacy	The platform shall support data minimisation, consent and customer privacy rights.
NFR-06	Accessibility	Core journeys shall conform to WCAG 2.1 AA principles.
NFR-07	Usability	A first-time user shall complete a standard transfer without external assistance.
NFR-08	Recoverability	Critical services shall support defined backup, recovery and continuity procedures.

8. Business Rules and Use Cases

Business rules

Rule ID	Business rule
BR-01	Only active customers with active accounts may initiate transfers.
BR-02	A transfer amount must be greater than zero and not exceed the available balance.
BR-03	Transactions above the customer threshold require step-up authentication.
BR-04	A newly added beneficiary may be subject to a cooling-off period or additional verification.
BR-05	Daily and per-transaction limits depend on customer type and risk profile.
BR-06	Every submitted transaction must receive a unique immutable reference number.
BR-07	Failed and reversed transactions must remain visible in history with an explanatory status.
BR-08	Support staff may view transaction status but may not access customer passwords or OTP values.

Online Banking Use-Case Overview



Detailed use case: Transfer funds

Field	Description
Primary actor	Retail customer
Preconditions	Customer authenticated; source account active; beneficiary available or valid recipient details entered.
Trigger	Customer selects Transfer Funds.
Main flow	Select source account; select/enter recipient; enter amount and reference; system validates; customer reviews; customer confirms with OTP; system posts or submits transfer; receipt and alert issued.
Alternative flows	Insufficient funds; invalid recipient; limit exceeded; OTP failed; integration timeout; duplicate submission detected.
Postconditions	Transaction status recorded; balances updated where applicable; audit log and receipt created.

9. User Stories and Product Backlog

Representative user stories

ID	User story	Acceptance criteria summary
US-01	As a customer, I want to sign in with MFA so that unauthorised users cannot access my accounts.	Given valid credentials, when the correct second factor is supplied, then the dashboard is displayed and the login event is logged.
US-02	As a customer, I want to view all account balances so that I understand my available funds.	Given an authenticated customer, when the dashboard loads, then all active accounts and available balances are displayed.
US-03	As a customer, I want to transfer money to a beneficiary so that I can make payments without visiting a branch.	Given a valid beneficiary and sufficient funds, when the customer confirms with OTP, then the transfer is submitted and a reference is displayed.
US-04	As a customer, I want to download a statement so that I can keep financial records.	Given a valid date range, when PDF is selected, then a readable statement is generated with opening and closing balances.
US-05	As a support agent, I want to search a transaction reference so that I can explain its status to the customer.	Given a valid reference, when searched, then the status, timestamps and non-sensitive details are displayed.

Prioritised backlog summary

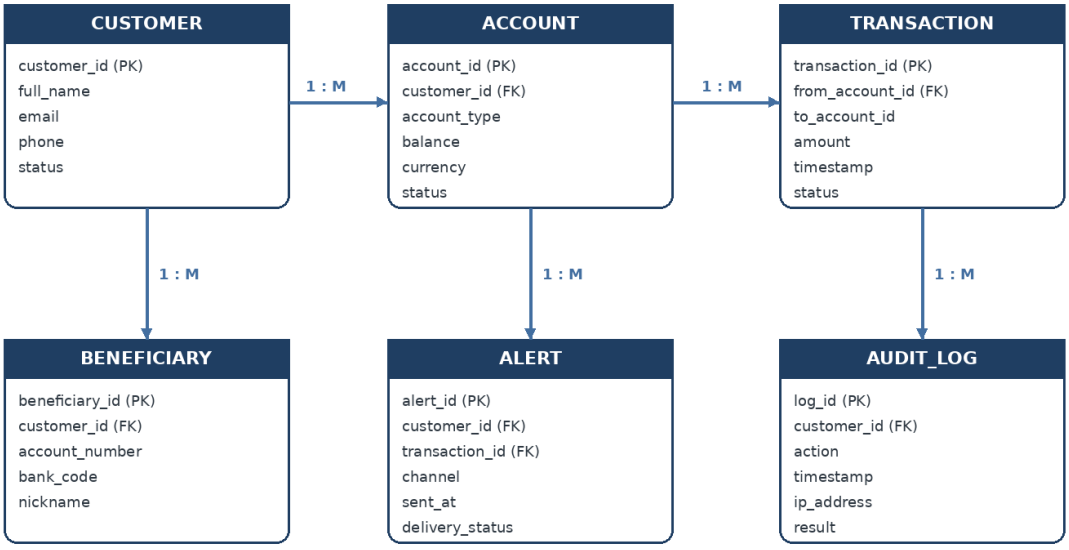
Rank	Story ID	Backlog item	Priority	Points	Target
1	US-01	MFA login	Must	8	Release 1
2	US-02	Account dashboard	Must	5	Release 1
3	US-03	Domestic transfer	Must	13	Release 1
4	US-06	Transfer validation and limits	Must	8	Release 1
5	US-07	Transaction receipt and reference	Must	5	Release 1
6	US-08	Audit logging	Must	8	Release 1
7	US-04	Statement download	Should	5	Release 2
8	US-09	Beneficiary management	Should	8	Release 2
9	US-10	Bill payment	Should	13	Release 2
10	US-11	Notification preferences	Could	5	Release 3

Agile delivery note

The backlog is prioritised using MoSCoW and estimated with indicative story points. Exact estimates would be agreed by the delivery team during refinement.

10. Data Model and Wireframes

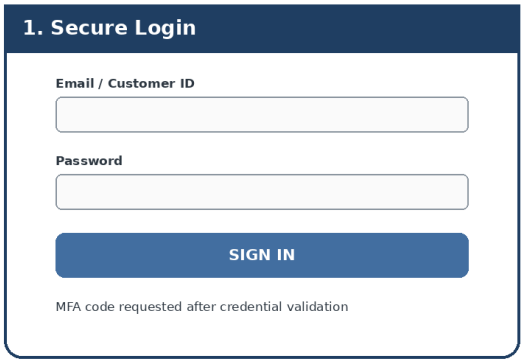
Simplified Data Model (ERD)

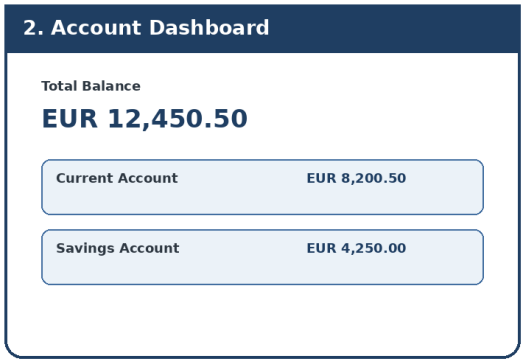


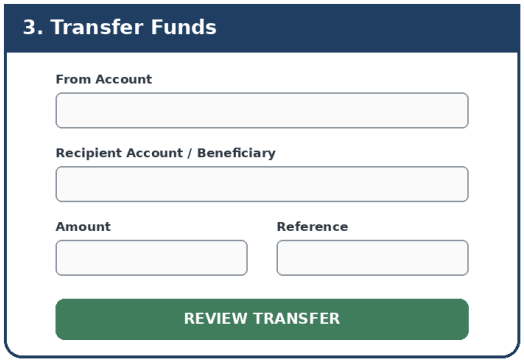
Data considerations

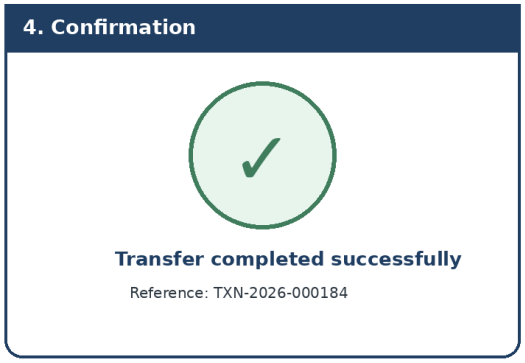
- Customer identity is separated from account and transaction records.
- Transactions reference source and destination details and retain status history.
- Audit logs capture sensitive actions without storing passwords or OTP values.
- Alerts link to customers and transactions to support delivery tracking.

Low-Fidelity Wireframes









Wireframe validation questions

- Can customers identify balances and primary actions immediately?
- Are validation messages clear and actionable?
- Does the review screen display recipient, amount, fees and reference before confirmation?
- Can the customer distinguish submitted, completed, pending, failed and reversed statuses?
- Are layouts understandable on both desktop and mobile devices?

11. UAT, Risk and Traceability

UAT sample

Test ID	Requirement	Scenario	Expected result	Priority
UAT-01	FR-01	Successful login with valid credentials and MFA	Dashboard displayed; authentication event logged	High
UAT-02	FR-01	Invalid OTP entered three times	Access denied; attempt logged; control action applied	High
UAT-03	FR-06/07	Transfer with valid beneficiary and sufficient funds	Transfer accepted; unique reference and receipt shown	High
UAT-04	FR-07	Transfer exceeds available balance	Submission blocked with clear message; no debit posted	High
UAT-05	FR-09	Completed transfer receipt	Receipt shows amount, recipient, date, status and reference	Medium
UAT-06	FR-10	Download statement for valid period	PDF generated with accurate transactions and balances	Medium
UAT-07	NFR-04	Review audit record for sensitive action	Actor, timestamp, action and result are recorded	High

Risk register summary

Risk ID	Risk	Rating	Mitigation
R-01	Account takeover	High	MFA, anomaly detection, session controls, customer alerts
R-02	Incorrect beneficiary details	High	Validation, confirmation screen, beneficiary verification
R-03	Duplicate transaction	High	Idempotency checks, duplicate warning and unique request ID
R-04	Core banking outage	High	Queue/retry strategy, clear pending status and reconciliation
R-05	Sensitive data exposure	High	Encryption, least privilege, masking and secure logging
R-06	Customer confusion	Medium	Usability testing, plain language and accessible error messages
R-07	Regulatory non-compliance	High	Compliance review, traceable controls and formal sign-off

Requirements traceability sample

Objective	Requirement	User story	UAT	Feature
OBJ-02	FR-06	US-03	UAT-03	Domestic transfer
OBJ-03	FR-01	US-01	UAT-01/UAT-02	Secure authentication
OBJ-03	NFR-04	US-08	UAT-07	Auditability
OBJ-04	FR-09	US-07	UAT-05	Receipt and status transparency
OBJ-05	FR-10	US-04	UAT-06	Self-service statements

12. Delivery Roadmap and Lessons Learned

Phase	Indicative focus	Key BA activities
Discovery	Problem validation, scope, stakeholders	Interviews, workshops, AS-IS analysis, objectives and assumptions
Definition	Requirements and solution options	BRD content, process models, business rules, NFRs and prioritisation
Release 1	Secure access, dashboard, transfers, audit	Refinement, acceptance criteria, traceability and UAT readiness
Release 2	Statements, beneficiaries, bill payments	Feedback analysis, backlog updates and change control
Release 3	Advanced alerts and service improvements	KPI review, optimisation and benefits tracking

Key lessons demonstrated

- Business Analysis connects objectives, user needs, controls, solution features and test evidence.
- Security and compliance must be built into requirements rather than added at the end.
- A requirement becomes more useful when it is specific, testable, prioritised and traceable.
- Process modelling makes customer pain points and future-state improvements easier to discuss.
- Wireframes help validate understanding before expensive development begins.
- A portfolio project is strongest when every claimed skill is supported by a visible deliverable.

Interview talking points

- Explain why branch dependency was selected as the business problem.
- Walk through the AS-IS and TO-BE transfer journey.
- Describe how FR-06, US-03 and UAT-03 connect through traceability.
- Discuss how MFA, limits, audit logs and duplicate controls reduce risk.
- Explain which features belong in Release 1 and why.
- Identify what additional stakeholder research would be required in a real bank.

Appendix: Evidence Package

The supporting workbook contains detailed, filterable evidence for the requirements, stakeholder register, user stories, prioritised product backlog, UAT test cases, risk register and traceability matrix. The GitHub-ready README explains the project structure and how a recruiter can review the deliverables.

Portfolio integrity
All deliverables should remain labelled as a simulated case study. During interviews, discuss what you created, why you made each decision and what you would validate differently with real users, bank data and regulatory experts.